

ANALYSIS OF DIGITAL TOOLS' USAGE ECONOMIC ASPECT IN THE HEALTHCARE SECTOR IN UKRAINE: GLOBAL CONTEXT AND WARTIME ADAPTATION

Kabanova Yelizaveta

Bachelor student of 124 System Analysis

Institute for Applied System Analysis

National Technical University of Ukraine "Igor Sikorsky Kyiv Polytechnic Institute"

ORCID ID: <https://orcid.org/0009-0001-2692-5066>

Melnychuk Viktoriia

PhD (Economics), Associate Professor

Department of Economic Cybernetics, Faculty of Management and Marketing

National Technical University of Ukraine "Igor Sikorsky Kyiv Polytechnic Institute"

ORCID <https://orcid.org/0000-0001-8246-4076>

Kuznietsova Nataliia

Doctor of Technical Sciences, Professor

Department of Mathematical Methods of System Analysis

Institute for Applied System Analysis

National Technical University of Ukraine "Igor Sikorsky Kyiv Polytechnic Institute"

ORCID ID <https://orcid.org/0000-0002-1662-1974>

Abstract

The article examines the economic and scientific potential of digital health in Ukraine, focusing on resilience and adaptation during crises. It compares Ukraine's system with global trends, analyzing standards, wartime challenges, and digitalization.

The study concludes that the sector has strong economic potential dependent on international cooperation, governance, data protection, and clear regulation.

Introduction

The rapid growth of cutting-edge technologies such as eHealth, telemedicine, AI, ML, and the Internet of Medical Things (IoMT) have become essential drivers of medical innovation, improving diagnostic accuracy, enhancing treatment efficiency, and reducing healthcare costs. International agencies such as the World Health Organization (WHO) [1] emphasize the role of digital innovation in promoting health, ensuring safety, and providing equitable access to medical services. Leading countries are actively leveraging digital technologies to overcome healthcare challenges, improve patient outcomes, and boost economies. From AI-powered diagnostics in the United States to Estonia's fully digital healthcare platform [4].

However, despite these advances, digital health strategies often face insufficient funding, fragmented resources, inconsistent standards, weak governance, and data-security concerns.

In Ukraine, the healthcare system relied mostly on conventional approaches but digital health has become increasingly important, especially during the COVID-19 pandemic and the full-scale war, which underscored the need for advanced analytical systems. Despite progress, Ukraine still faces challenges including limited funding, regulatory gaps, public distrust, and unequal technological access.

Materials and methods

Digital medicine is an interdisciplinary field that involves the use of digital technologies and concepts for diagnostic, therapeutic, and scientific purposes. The related products are based on research, development and clinical experimentation with an emphasis on evidence-based use of digital tools [2]. Currently, the spectrum of digital health technologies has expanded to include big data, telemedicine, Artificial Intelligence (AI), Machine Learning (ML), Internet of Things (IoT), and many others technologies. [5].

These technologies are especially useful for remote medical facilities and institutions, and in cases of transportation of medical products, where there are high logistical risks. They manage the situation, warn about the emergence of problems that may threaten the safety of patients or damage medical drugs and equipment, and are even able to monitor the disinfection of the hands of medical personnel, which significantly reduces the risk of outbreaks of infections.

According to the World Health Organization, countries spent \$9 trillion on healthcare in 2020 – 11% of the global gross domestic product – while a significant portion of expenditure on healthcare is wasteful due to systemic inefficiency [7]. AI systems are already reducing such inefficiencies: for instance, Kheiron Medical uses AI to assess mammograms and detect cancer without requiring a second radiologist, while Methinks accelerates stroke diagnosis by analyzing preliminary CT scans [11].

The economic impact of digital healthcare also extends to hospital management. In a study conducted in a hospital operating in Ankara implementing HIMSS (Healthcare Information and Management Systems Society), it was determined that the primary costs of economic benefits in the digital hospital were paper and printing device depreciation expenses, toner fees, archive depreciation costs, and labor expenses [10].

The digital transformation of healthcare also involves digital forms of patient communication with medical staff. Chatbots optimize the process of the service, reduces the burden on medical institutions and saves time for doctors and medical staff, as patients are able to describe their symptoms in a calm manner without the pressure of limited time at a doctor's appointment.

In contrast, in Ukraine, digital services of this level are just beginning to be introduced in large clinics and leading diagnostic centers. Paper medical records in Ukraine still have greater legal weight than digital ones, and most medical institutions use both paper and electronic documentation. This hinders the development of digital media, as electronic records are often considered only an alternative [15].

Digital medicine has significantly accelerated during the COVID-19 era. Hospitals reported that although the number of in-person visits declined dramatically, the use of medical services peaked through mobile applications, websites, and chatbots [9]. Similarly, the acceleration of telemedicine in Ukraine was primarily due to the pandemic and later the war [8].

Despite the barrier, Ukraine achieved progress in the development of the Electronic Health Care System. Today, eHealth is one of the largest electronic systems in Ukraine, with its functionality constantly updated. The exchange of necessary medical data is made via an open software interface (API), connecting the central database with disease-specific registries, state registers of facilities, pharmaceuticals, treatment standards, and healthcare professionals [15]. These registries support predictive modeling, resource allocation, and wartime healthcare logistics. A key component is the integrated electronic medical record (EMR), which consolidates patient history and individual disease characteristics, enabling doctors to assess medical, psychological, and pedagogical needs [26].

Global surveys and research also point to noticeable market growth. According to Statista, revenue in the digital health market is projected to reach USD 258.25 billion in 2029 with a CAGR of 6.88% (Figure 1). In comparison, Ukraine's digital health market in Ukraine is projected to reach USD 209.45 million in 2029 with a CAGR of 1.73% (Figure 2), which is significantly lower than the global average [6]. Still, the projected dynamics confirm that digital health remains one of the few growing segments of the Ukrainian healthcare economy even under war conditions.

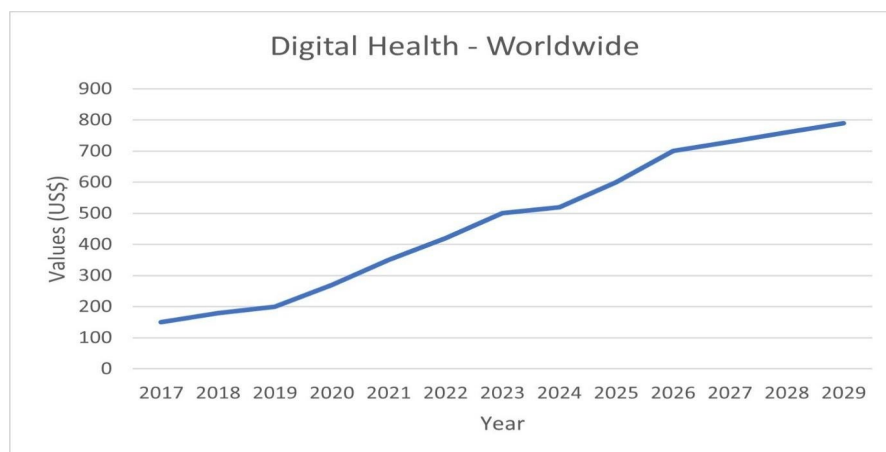


Figure 1. Revenue digital health market worldwide (*Author's development*)

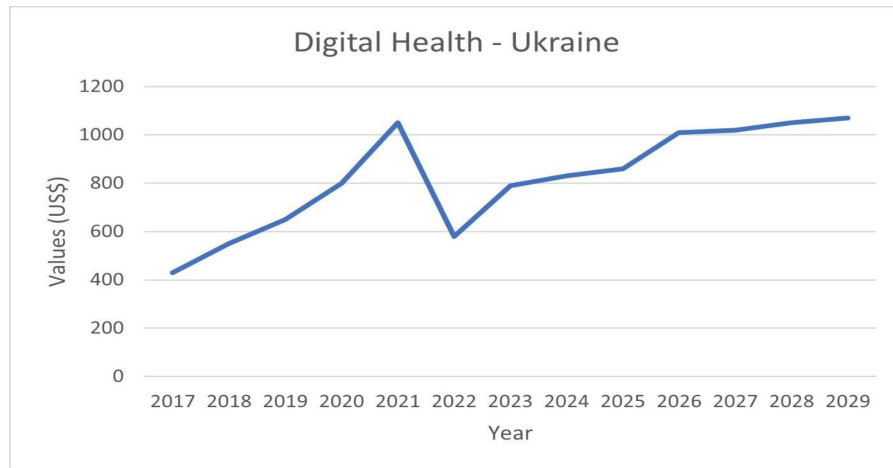


Figure 2. Revenue digital health market in Ukraine (*Author's development*)

Ukraine's government health expenditures as a share of GDP remain lower than those in the USA, the UK, EU, and global average since the Russian invasion in 2014 (Figure 3). This highlights a significant pressure caused by the war.

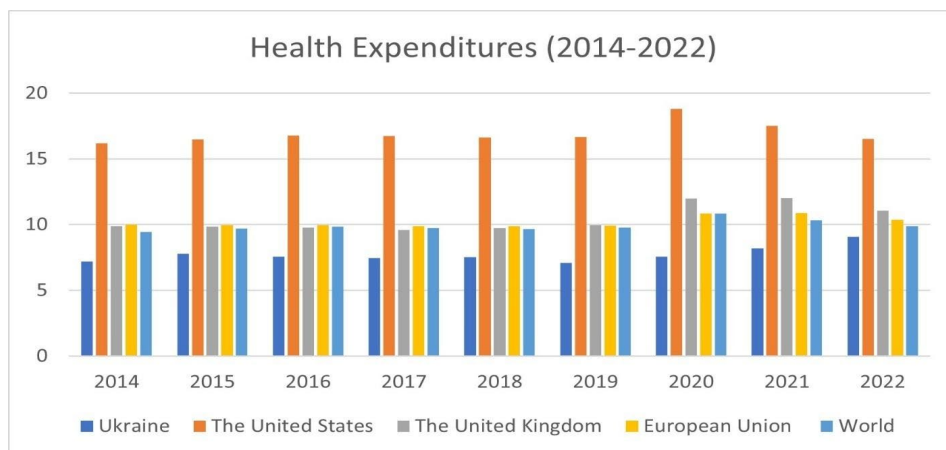


Figure 3. Global Health Expenditures by different regions worldwide (*Author's development*)

Digital health has also provided new career opportunities. Professionals with certification in health informatics are in high demand, as hospitals and healthcare systems seek experts who can manage data systems, analyze health data, and ensure the successful implementation of digital tools [4]. Ukraine has not established health informatics as an independent field of study yet, but it continues to introduce new innovations that accelerate conventional healthcare practices. Like that, "Nursing Staff Workplace" system enables nurses to manage electronic medical records and record

patients' treatments, thereby redistributing doctors' workload and increasing the nursing staff's role in care [13].

Estonia's e-Health system connects all healthcare providers through a centralized digital platform, while Singapore has invested in "smart hospitals" to monitor patients and manage hospital operations more efficiently by leveraging big data and AI [4]. Similarly, Ukraine's system eZdorovya now serves over 17 million patients, enabling electronic health records along with digital tools for healthcare providers, administrators, and pharmacists [14].

Despite developments, more than 40% of Ukrainians still have a lower than basic level of digital literacy. This created difficulties in understanding and using digital health solutions for certain social groups, such as minorities, the elderly, and those in low-income or rural areas. Nevertheless, Ukraine managed to preserve its digital transformation. Except eHealth becoming one of the largest electronic systems in Ukraine, Telemedicine services received legal foundation through the adoption of the Strategy for the Development of Telemedicine in Ukraine and the Law of Ukraine [17, 18]. The introduction of such regulations benefits economically, reducing travel costs through remote consultations, and also ensures greater safety for the nation.

Limited financial resources delay the introduction of new technologies, especially in frontline areas due to the low population and difficult access to infrastructure. Additionally, the Internet speed and convergence is still below that of EU countries [22].

To address these challenges, the government of Ukraine introduced the Strategy for the Development of the Electronic Communications Sector until 2030, aimed at creating opportunities for connecting to the Internet for the maximum number of citizens, public institutions, enterprises, organizations, and comprehensive use of the opportunities that arise for users of the global network [23]. These changes are expected to accelerate the development of digital health by ensuring more reliable Internet connections and access to health services across the country.

Whereas the European Union has set an ambitious goal of providing 100% of citizens with access to their electronic health records by 2030, Ukraine is actively taking part in initiatives with the European Health Data System (EHDS) [12, 16]. The governance seeks deeper integration into the European Research Area to drive economic growth and innovations. Following these goals, the Strategy for the Digital Development of Innovative Activity of Ukraine until 2030 aims to create new opportunities in research and innovations [23].

The Ukrainian government has also set the task of the phased introduction of e-health tools through legal principles. The Law of Ukraine "On State Financial Guarantees of Health Care Services" and the Resolution of the Cabinet of Ministers of Ukraine "Some Issues of the Electronic Health Care System" ensure patient access to essential, high-quality medical services and medicines funded by the State Budget within the medical guarantees program [12]. These acts revise financial principles and establish the e-health system, which enhances interaction among all stakeholders — government, providers, medical professionals, and patients — through the cost-effective and secure use of information and communication technologies.

Another key aspect of implementing digital healthcare is personal data safety. Ukraine's digital healthcare system meets both international standards and Ukrainian standards [15]. Data protection in the central database is regulated by the Law "On Information Protection in Information and Communication Systems", data processing complies with the Law "On Personal Data Protection" [12]. With support from international experts, the system follows global best practices to prevent cyberattacks and unauthorized access.

To strengthen data security, the government adopted the "Concept of the Electronic Healthcare System", which defines priorities for e-health development through 2025. These include improving service requirements, enhancing electronic identification and authentication, unifying data exchange standards, maintaining central registers, regulating access to medical data, ensuring interoperability with other systems, and aligning national standards with international ones.

At the same time, legislation, both Ukrainian and international, often fails to keep pace with technological changes. Today, most countries in the world do not have clearly developed legal norms that would regulate the use of digital services in the medical industry, which creates legal uncertainty for service providers and users [15]. The issue is also in determining the responsible party in case of errors due to software failures [10].

The ongoing war also poses serious threats to Ukraine's digital healthcare system, making the task of table functioning of the entire healthcare system under martial law a priority. Since February 24, 2022, dozens of health facilities and health workers have been attacked and displaced, and continuous attacks on energy systems have jeopardized system stability. Due to the displacement of Ukrainians, the number of users increased in the west of the country, causing errors in some medical information systems [3]. According to Statista, average revenue from digital healthcare dropped by almost one-third in 2022 [6]. The paramount aim is to enable connections between experts from different countries in real-time in order to receive professional recommendations and to provide all those who need it with medical care regardless of the distance and request time.

However, Ukraine is actively developing digital medicine, adapting it to the challenges of wartime conditions. On January 1, 2025, a new electronic system for assessing daily functioning was launched, transferring medical and social expert commissions to a fully digital format. Its goal is to reduce bureaucracy and streamline interactions with healthcare providers. The system, built on the Diia.Engine platform, uses low-code components that accelerate development and simplify maintenance, as Diia.Engine is a standardized state solution. The platform ensures a transparent and secure assessment process, and the Ministry of Health plans to integrate it with the national e-health system and other state registries during 2025.

While global practices clearly demonstrate the benefits of digital transformation, many Ukrainians still face barriers in access to healthcare. According to the Health Index Ukraine 2023 survey, nearly half of respondents who did not seek medical care

(49.3%) believed they could treat themselves, and 23.4% hoped that the disease would pass without medical care. Financial concerns (13.9%) and queues in facilities (13.2%) were also notable barriers. Additionally, 8.5% reported distrust of medical staff, and 4.1% cited obstacles related to the full-scale war [20].

Older adults (60+) were more likely to refrain from seeking care due to treatment costs and distrust in providers. In 2023, people aged 65+ accounted for 20.1% of the population. Urban residents more frequently complained about queues (15.8% vs. 7.4%), while rural residents faced transportation barriers (7.3% vs. 2.0%). Given that roughly 30% of the population lives in rural areas, accessibility challenges remain significant [21]. The results of this survey highlight the main drawbacks in the traditional healthcare system: unaffordable prices, poor management, and a lack of trust in medical service among Ukrainians.

Conclusion

Digital healthcare offers substantial economic benefits by reducing administrative costs, improving resource allocation, and enhancing diagnostic accuracy. It has become essential for maintaining access to medical services, underscoring the need for strong data protection and clear regulatory frameworks.

Although Ukraine lags behind global leaders, its accelerated digitalisation during COVID-19 and the full-scale war through the eZdorovya platform, medical registries, and telemedicine legislation. However, persistent barriers, such as limited funding, low digital literacy, and reliance on paper-based documentation remain.

The study shows that Ukraine's future digital health development will depend on deeper international cooperation, clearer regulation, stronger data protection, and continued support for innovation.

References

1. World Health Organization. Global strategy on digital health 2020–2025. Geneva, 2021. URL:<https://apps.who.int/iris/bitstream/handle/10665/344249/9789240020924-eng.pdf> (last accessed: 24.09.2025).

2. Mumtaz H., Riaz M. H., Wajid H., Saqib M., Zeeshan M. H., Khan S. E., Chauhan Y. R., Sohail H., Vohra L. I. Current challenges and potential solutions to the use of digital health technologies in evidence generation: a narrative review. *Frontiers in Digital Health*. 2023. DOI: <https://doi.org/10.3389/fdgth.2023.1203945>.
3. Malakhov K. Insight into the Digital Health System of Ukraine (eHealth): Trends, Definitions, Standards, and Legislative Revisions. *International Journal of Telerehabilitation*. 2023. Vol. 15, No. 2. DOI: <https://doi.org/10.5195/ijt.2023.6599>.
4. Betty D. Global Experience: How Other Countries Are Using Digital Technologies in Healthcare. Coachorg. 2024. URL: <https://www.coachorg.com/global-experience-how-other-countries-are-using-digital-technologies-in-healthcare/> (last accessed: 24.09.2025).
5. Wai A., Torkamani A., Butte A. J., Glicksberg B. S., Schuller B. W., Rodríguez B., Shu D., Bates D. W., Schaden E., Peng H., Willschke H., Car J., Rahimi K., Celi L. A., Banach M., Kletečka-Pulker M., Kimberger O., Eils R., Shariful M. The promise of digital healthcare technologies. *Frontiers in Public Health*. 2023. Vol. 11, No. 1196596. DOI: <https://doi.org/10.3389/fpubh.2023.1196596>.
6. Statista. Digital Health – Worldwide. URL: <https://www.statista.com/outlook/hmo/digital-health/worldwide?currency=USD> (last accessed: 24.09.2025).
7. World Economic Forum. How digital healthcare tools cut costs and boost outcomes. 2024. URL: <https://www.weforum.org/impact/how-digital-healthcare-tools-cut-costs-boost-outcomes/> (last accessed: 24.09.2025).
8. Poberezhets V., Demchuk A., Mostovoy Y. How Russian-Ukrainian War Changed the Usage of Telemedicine: A Questionnaire-Based Study in Ukraine. *Ankara Medical Journal*. 2022. Vol. 22, No. 3. P. 305–318. DOI: <https://doi.org/10.5505/amj.2022.08455>.
9. Getachew E., Adebeta T., Muzazu S. G. Y., Charlie L., Said B., Tesfahunei H. A., Wanjiru C. L., Acam J., Kajogoo V. D., Solomon S., Atim M. G., Manyazewal T.

Digital health in the era of COVID-19: Reshaping the next generation of healthcare. *Frontiers in Public Health*. 2023. Vol. 11, No. 942703. DOI: <https://doi.org/10.3389/fpubh.2023.942703>.

10. Torun N. Economic Effects of Digitalization: Health Sector Example. *Anemon Muş Alparslan Üniversitesi Sosyal Bilimler Dergisi*. 2024. Vol. 12, No. 2. P. 717–724. DOI: <https://doi.org/10.18506/anemon.1507704>.

11. European Parliament. The digitalisation of healthcare. 2021. URL: [https://www.europarl.europa.eu/RegData/etudes/BRIE/2021/690548/EPRS_BRI\(2021\)690548_EN.pdf](https://www.europarl.europa.eu/RegData/etudes/BRIE/2021/690548/EPRS_BRI(2021)690548_EN.pdf) (last accessed: 24.09.2025).

12. European Commission. Digital Decade 2024: eHealth Indicator Study. 2024. URL: <https://digital-strategy.ec.europa.eu/en/library/digital-decade-2024-ehealth-indicator-study> (last accessed: 24.09.2025).

13. Myronova G. DIGITALIZATION OF HEALTHCARE IN UKRAINE: LEGAL SUPPORT OF PUBLIC AND PRIVATE INTERESTS. *Entrepreneurship, Economy and Law*. 2022. No. 3. P. 40–47. DOI: <https://doi.org/10.32849/2663-5313/2022.3.05>

14. Observatory of Public Sector Innovation. The Public Private Collaboration model, eZdorovya, and the transformation of the healthcare system in Ukraine. 2022. URL: <https://oecd-opsi.org/innovations/the-public-private-collabroation-model-ezdorovya-and-the-transformation-of-the-healthcare-system-in-ukraine/> (last accessed: 24.09.2025).

15. Valchuk M. S. Digitalization of the healthcare sector in Ukraine on the way to ensuring client and patient orientation. *Central Ukrainian Journal of Law and Public Administration (CUJ)*. 2024. No. 3. P. 14–22. DOI: <https://doi.org/10.32782/cuj-2024-3-2>

16. Ukrainian World Congress. EU to connect Ukraine to European health data system starting in 2026. 2025. URL: <https://www.ukrainianworldcongress.org/eu-to-connect-ukraine-to-european-health-data-system-starting-in-2026/> (last accessed: 24.09.2025).

17. The Government of Ukraine. On approval of the Strategy for the Development of Telemedicine in Ukraine. 2023. URL: <https://www.kmu.gov.ua/npas/pro-skhvalennia-stratehii-rozbudovy-telemetrytsyny-v-ukraini-i140723-625> (last accessed: 24.09.2025).
18. Verkhovna Rada of Ukraine. Law of Ukraine No. 3301-IX “On Rehabilitation in the Field of Healthcare”. 2021. URL: <https://zakon.rada.gov.ua/go/1053-20> (last accessed: 24.09.2025).
19. Market.us. Medical Technology and Innovation Statistics. 2025. URL: <https://media.market.us/medical-technology-and-innovation-statistics/> (last accessed: 24.09.2025).
20. National Study. Health Index Ukraine Report. 2023. URL: https://healthindex.com.ua/HI_Report_UA_2023_EN.pdf (last accessed: 24.09.2025).
21. State Statistics Service of Ukraine. Statistical Yearbook of Ukraine for 2022. 2023. URL: https://www.ukrstat.gov.ua/druk/publicat/kat_u/2023/zb/11/year_23_e.pdf (last accessed: 24.09.2025).
22. Cabinet of Ministers of Ukraine. On approval of the Concept for the development of eHealth: Decree No. 1671-r, 28 December 2020. URL: <https://zakon.rada.gov.ua/go/1671-2020-%D1%80> (last accessed: 24.09.2025).
23. Ministry of Health of Ukraine. Digital transformation of medical and social expertise: from 2025 all processes will be in the electronic system. 2025. URL: <https://moz.gov.ua/uk/news/cifrova-transformaciya-mediko-socialnoyi-ekspertizi-z-2025-roku-vsi-procesi-v-elektronnij-sistemi> (last accessed: 24.09.2025).